

PROGRAM

SENDER

```
#include <stdio.h>
#include <sys/ipc.h>
#include <sys/msg.h>

// Step 1: Create structure
struct msg_buffer {
    long msg_type;
    char msg_text[100];
} message;

int main() {
    key_t key;
    int msgid;

    // Step 2: Generate Key
    key = ftok("progfile", 65);

    // Step 3: Create Message Queue
    msgid = msgget(key, 0666 | IPC_CREAT);
    message.msg_type = 1;

    // Step 5: Get message
    printf("Write Data : ");
    fgets(message.msg_text, 100, stdin);

    // Step 6: Send message
    msgsnd(msgid, &message, sizeof(message), 0);

    // Step 7: Print
    printf("Data send is : %s\n", message.msg_text);

    return 0;
}
```

RECEIVER

```
#include <stdio.h>
#include <sys/ipc.h>
#include <sys/msg.h>

struct msg_buffer {
    long msg_type;
    char msg_text[100];
} message;

int main() {
    key_t key;
    int msgid;

    // Step 2: Generate Key
    key = ftok("progfile", 65);

    // Step 3: Open Queue
```

```
msgid = msgget(key, 0666 | IPC_CREAT);

// Step 4: Receive message
msgrcv(msgid, &message, sizeof(message), 1, 0);

// Step 5: Print
printf("Data Received is : %s\n", message.msg_text);

// Step 6: Remove Queue
msgctl(msgid, IPC_RMID, NULL);

return 0;
}
// Note: Create a dummy file named 'progfile' in the same folder before running.
```

OUTPUT

```
blackie@blackie-pc:~/GEC_PKD_IT/S4/OS_LAB/EXPERIMENT_4$ gcc 4c_sender.c
blackie@blackie-pc:~/GEC_PKD_IT/S4/OS_LAB/EXPERIMENT_4$ ./a.out
Write Data : Hi
Data send is : Hi

blackie@blackie-pc:~/GEC_PKD_IT/S4/OS_LAB/EXPERIMENT_4$ gcc 4c_receiver.c
blackie@blackie-pc:~/GEC_PKD_IT/S4/OS_LAB/EXPERIMENT_4$ ./a.out
Data Received is : Hi
```